Omar Guettat *Embedded Software Engineer | Firmware, IoT & Embedded Linux*

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PROFILE

Industrial Computer Engineering graduate (ENET'Com, 2026) specialized in embedded firmware development and embedded Linux systems. Hands-on experience designing and validating a 5-node distributed embedded platform (Raspberry Pi, Arduino) over CAN, LIN, and Ethernet/SOME-IP, following a structured V-model methodology from requirements documentation through test validation. Academic grounding in RTOS principles, networking, and industrial fieldbus protocols (Modbus, I2C, SPI). Active IEEE member, open to full-time embedded software engineering roles.

EDUCATION

Engineering Degree in Industrial Computer Engineering 2023 – 2026 | Sfax, Tunisia
National School of Electronics and Telecommunications of Sfax (ENET'Com)
Coursework: Test & Validation (ISTQB principles), RTOS, Computer Networks (CCNA), Fieldbus Protocols (Modbus, I2C, SPI), IoT Networks, Smart Grid & Renewable Energy Systems, Signal Processing, Control Systems (MATLAB/Simulink)

Preparatory Cycle in Physics-Technique 2021 – 2023 | Sfax, Tunisia
Preparatory Institute for Engineering Studies of Sfax

SKILLS

Embedded Systems & Firmware
Embedded C/C++ (bare-metal and RTOS), microcontroller firmware, register-level peripheral configuration, HAL/driver architecture, RTOS; Arduino, ESP32, Raspberry Pi

Protocols
CAN (ISO 11898), LIN 2.0, SOME/IP, I2C, SPI, UART, RS232, Modbus, MQTT

Programming Languages & Tools
Python, C, C++, Java, Git, CMake (basic), PlatformIO

Embedded Linux
SocketCAN, device tree overlays, systemd services, serial driver debugging (termios), shell scripting

Test & Validation
V-model, requirements documentation, test case design, test automation, ISTQB principles, logic analyzer, UART diagnostics

Languages
Arabic, French, English

PROFESSIONAL EXPERIENCE

Embedded Systems Engineer Intern (PFE) 02/2026 – 06/2026 | Sfax, Tunisia
KPIT (Internship)
- Designed and built a 5-node distributed embedded system with a 4-layer architecture (Application, Service, Driver, HAL) over CAN (ISO 11898, 500 kbit/s) and LIN (LIN 2.0, 19.2 kbit/s)
- Followed a V-model: wrote requirements and specifications first, then validated with 10 documented test cases (including boundary/negative tests), achieving a 100% pass rate
- Implemented a LIN master scheduler with low-level UART register programming; fixed a Linux serial-driver edge case (false LIN BREAK detection) via termios
- Designed a custom CRC-8 (SAE J1850) integrity scheme for CAN and a SOME/IP publish-subscribe service with Service Discovery and WebSocket dashboard bridging
- Diagnosed bus-level faults with a logic analyzer and UART diagnostics; built a fault-injection dashboard automating 5 ECU fault scenarios
- Reverse-engineered two undocumented automotive components with only a multimeter, designing a voltage-divider circuit to interface one panel to an Arduino ADC

Design and Implementation of a Local RAG- based Chatbot 07/2025 – 08/2025 | Megrine, Tunisia
Sagemcom (Internship)
- Designed and implemented a local RAG-based chatbot (LangChain, FAISS, BM25, Sentence Transformers) with a FastAPI backend and React frontend to improve knowledge transfer within the Quality & Environment department

PROJECTS

Industrial Monitoring System & Smart HVAC Control – IoT Project 10/2025 – 12/2025
- Built an ESP32-based system for real-time industrial gas and environmental monitoring using 4 sensors tracking 6 parameters (CO, CO_2, PM10, PM2.5, temperature, humidity), transmitting sensor data via MQTT, with resource-conscious firmware design for the microcontroller's limited RAM/flash
- Implemented automatic HVAC control based on exposure threshold detection (WHO/OSHA/NIOSH), with Python backend for data storage and risk prediction

Industrial Production Management with Raspberry Pi- End-of-year project 10/2024 – 03/2025
- Developed a Raspberry Pi-based signal processing system for preventive maintenance, with a web interface (React, Node.js) for remote performance visualization

ORGANISATIONS

IEEE ENET'Com Student Branch 02/2025 – 02/2026
Treasurer
Managed the branch budget and led sponsorship negotiations and income

IAS ENET'Com Student Branch Chapter 02/2024 – 02/2025
Chair
- Led a team organizing technical and non-technical events
- Organized 50+ workshops and activities across both roles
- Designed and conducted two multi-session IoT bootcamps (2025, ongoing)